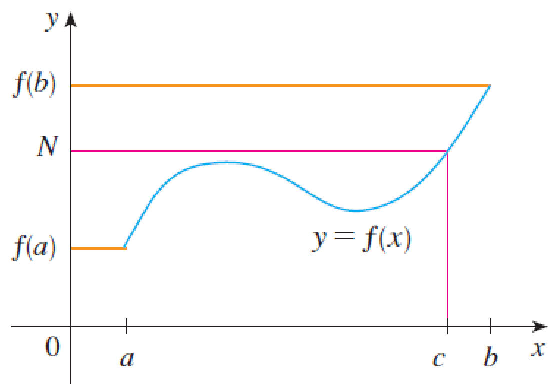
2.4.2 The Intermediate Value Theorem

Question to the Class: Suppose f is a continuous function on the set of real numbers. Suppose you know that $f(3) = -4$ and $f(10) = 2$. Must this function have a zero (i.e. an x value such that $f(x) = 0$)?

Note: The key here is to think about what the graph will look like. The next theorem addresses this question.

Theorem (The Intermediate Value Theorem):
Suppose that f is continuous on the closed interval $[a, b]$ and let N be any number between $f(a)$ and $f(b)$ where $f(a) \neq f(b)$. Then, there exists a number c in (a, b) such that $f(c) = N$.

Proof: The proof is beyond the scope of this course and is not included in the text, but the picture below illustrates the idea.



Example 1: Suppose that Bob and Alice each take a 9 hour trip (during the same time interval). The temperature that Bob is experiencing at time t (in hours) is given by the continuous function $B(t)$. Similarly, the temperature for Alice is given by the continuous function $A(t)$ (both functions are continuous on $[0,9]$). Suppose that you are told $B(0) = A(9) = -15$ and $B(9) = A(0) = 100$. Prove there is a time c such that $B(c) = A(c)$.

Let $f(t) = B(t) - A(t)$. By a theorem from the previous section f is continuous on $[0,9]$.
Moreover, we know that $f(0) = B(0) - A(0) = -15 - 100 = -115$ and $f(9) = B(9) - A(9) = 100 - (-15) = 115$.
By the intermediate value theorem there must exist a $c \in (0,9)$ such that $f(c) = 0$.
This implies that $B(c) = A(c)$ as desired.

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Note: This solves one of the intro problems from the first day of class!

Example 2: Show that there is a root (i.e. solution) to the equation: $4x^3 - 6x^2 + 3x - 2 = 0$.

We consider the function $p(x) = 4x^3 - 6x^2 + 3x - 2$ which we know is continuous for all real numbers.
We note that $p(0) = -2$ and $p(2) = 4 \cdot 8 - 6 \cdot 4 + 3 \cdot 2 - 2 = 12$.
By the intermediate value theorem there must exist a $c \in (0, 2)$ such that $p(c) = 0$.
This immediately implies that the equation has a root (In fact, we have proven there is a root between 0 and 2).
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Note: Really, for the above example we just need one value of x that produces a negative range value and one value of x that produces a positive range value. So, many proofs are possible.

Example 3: Show that the graph of function $f(x) = \sin(x) - x$ obtains a *y*-value of -5 at least once.

Since $g(x) = \sin(x)$ and $h(x) = x$ are continuous for all real numbers, we know that f is continuous because f is the difference of these two functions.
We note that $f(0) = \sin(0) - x = 0$ and $f(2\pi) = \sin(2\pi) - 2\pi = -2\pi$

By the intermediate value theorem there must exist a $c \in (0, 2\pi)$ such that $f(c) = -5$.
This immediately implies that the function obtains a *y*-value of -5 at least once in the interval $(0, 2\pi)$.
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